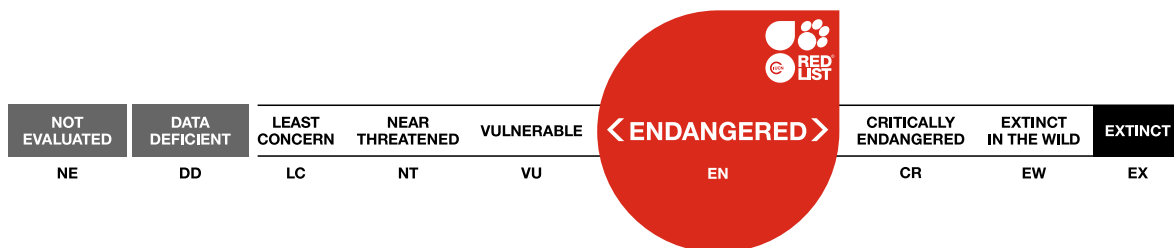


Brachiopsilus dianthus, Pink Handfish

Assessment by: Edgar, G., Last, P.R. & Stuart-Smith, R.



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Actinopterygii	Lophiiformes	Brachionichthyidae

Scientific Name: *Brachiopsilus dianthus* Last & Gledhill, 2009

Common Name(s):

- English: Pink Handfish

Taxonomic Source(s):

Eschmeyer, W.N., Fricke, R., and Ven der Laan, R. (eds.). 2017. Catalog of Fishes: genera, species, references. Updated 01 December 2017. Available at: <http://researcharchive.calacademy.org/research/ichthyology/catalog/fishcatmain.asp>.

Assessment Information

Red List Category & Criteria: Endangered B1ab(v)+2ab(v) [ver 3.1](#)

Year Published: 2020

Date Assessed: April 30, 2018

Justification:

This shallow, demersal species is relatively poorly known and is inferred to be restricted to a small area in southeastern Tasmania, Australia. The estimated extent of occurrence (EOO) is 1,262 km² and area of occupancy (AOO) is 12 km². The three subpopulations from which it is known are considered severely fragmented due to the low dispersal capabilities of this species. Extensive surveys targeting handfish species have been carried out throughout this area in the shallow habitats that this species occupies, but no individuals have been found, and the last record was confirmed in 1999. Other better-known, shallow handfish species that are endemic to this same area are highly restricted in range and have experienced documented serious declines. Serious impacts from historical scallop dredging that took this species as bycatch and destroyed habitat are likely to have occurred. Therefore, it is listed as Endangered B1ab(v)+2ab(v). Further research and monitoring is needed on the population trends and biology of this species.

Geographic Range

Range Description:

This species is distributed in southeastern Tasmania, Australia from three localities: the Tasman Peninsula, the mouth of the Huon River and off Dennes Point in the D'Entrecasteaux Channel. Its depth range is about 15-38 metres (Last and Gledhill 2009). Many handfish species have highly restricted ranges and subpopulations are severely fragmented due to their low dispersal capabilities; therefore, despite being known from a low number of records, it is inferred that this species is restricted to this area of southeastern Tasmania. Based on a minimum bounding polygon drawn around the three known points, the extent of occurrence (EOO) is estimated as 1,262 km². Since handfish subpopulations are

typically restricted to areas less than 100 metres in diameter (Edgar *et al.* 2015, Edgar *et al.* 2017), the area of occupancy (AOO) is estimated as the area of three 2 x 2 km grid cells, which is 12 km².

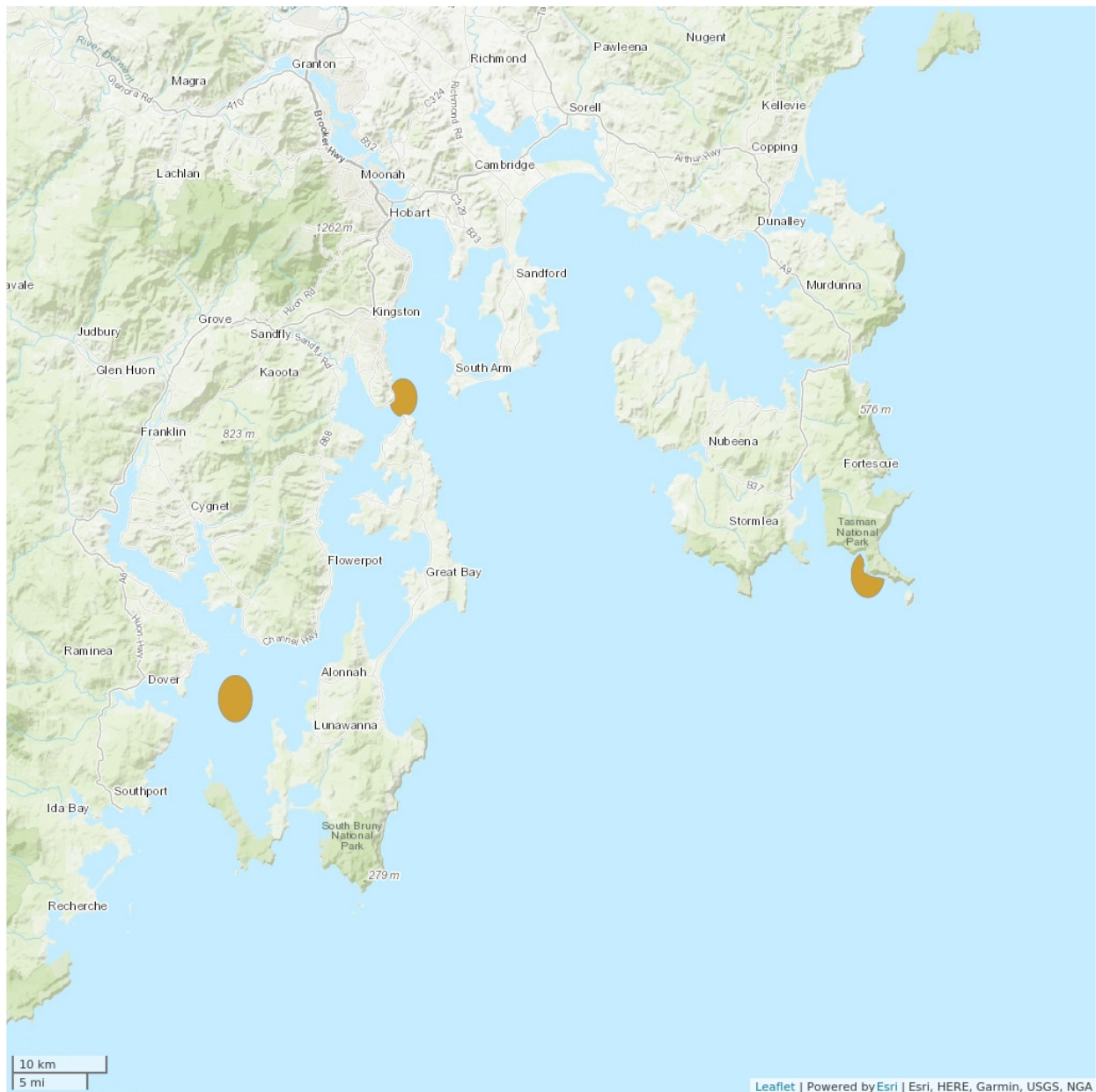
Country Occurrence:

Native, Extant (resident): Australia (Tasmania)

FAO Marine Fishing Areas:

Native: Indian Ocean - eastern

Distribution Map

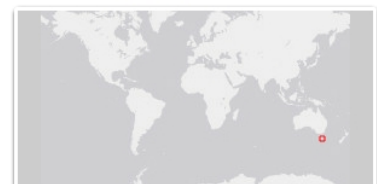


Legend

● EXTANT (RESIDENT)

Compiled by:

International Union for Conservation of Nature 2019



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

This species is known from only five specimens, the most recent record is from 1999 and it has not been observed for the past 20 years despite considerable sampling effort by commercial, recreational and scientific entities across its known range (Last and Gledhill 2009, Edgar *et al.* 2017). Due to their low dispersal capabilities and relatively low reproductive rate, handfish species occur in isolated subpopulations (T. Lynch pers. comm. 2018); therefore, this species is considered severely fragmented since more than half of the individuals are in small and isolated patches. In recent decades, extensive underwater surveys targeting shallow handfish species have been carried out across south and eastern Tasmania, but no individuals of this species were found (Edgar *et al.* 2017). For example, a total of 100 hours were spent by 19 experienced divers at 22 sites across southern Tasmania from February to June 2015 to specifically search for Ziebell's Handfish (*Brachiopsilus ziebelli*) and the Red Handfish (*Thymichthys politus*), which are similar in ecology and distribution to this species (Edgar *et al.* 2015, Edgar *et al.* 2017). Survey work was repeated in 2017 and 2018 off the Tasman Peninsula, but no individuals were recorded (R. Stuart-Smith pers. comm. 2018). Based on well-documented declines in the Red Handfish, Ziebell's Handfish and the Spotted Handfish, it is inferred that this species has also undergone declines in the number of mature individuals (Last and Gledhill 2009, Edgar *et al.* 2017).

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

This demersal species inhabits shallow, inshore soft bottoms probably near or on hard substrates (rocky/rubble) amongst thick coverage of macroinvertebrates (R. Stuart-Smith and P. Last pers. comm. 2018). Its maximum total length is 13.6 cm (Last and Gledhill 2009). Handfish species do not have a pelagic larval stage, and dispersal of propagules is very limited (Edgar *et al.* 2017). Females lay small clusters of large eggs attached to the substrate by filaments and some level of parental care is provided until they hatch (Last and Gledhill 2009).

Systems: Marine

Use and Trade (see Appendix for additional information)

This species is not utilized. It was taken as bycatch in the historical scallop fishery and may also be caught in lobster traps currently (R. Stuart-Smith and P. Last pers. comm. 2018).

Threats (see Appendix for additional information)

Other shallow handfish species (e.g., *Brachionichthys hirsutus*) that occur in the same region as this species have experienced significant population declines due to their very limited dispersal capabilities, small subpopulations and impacts from invasive species, impacts from historical scallop dredging, habitat loss and pollution (Edgar *et al.* 2017). A large scallop fishery that was active in southeastern Tasmania from the 1800s until its collapse in the D'Entrecasteaux Channel by 1967 probably caused population declines due to habitat destruction and direct take of individuals as bycatch (Edgar and Samson 2004, Last and Gledhill 2009, R. Stuart-Smith and P. Last pers. comm. 2018).

Conservation Actions (see Appendix for additional information)

There are no protected areas or species-specific conservation measures. Further research is urgently

needed on this species, as little is known on its biology, distribution and population, and there is a high potential that declines may have occurred similar to other handfish species (Last and Gledhill 2009, Edgar *et al.* 2017).

Credits

Assessor(s): Edgar, G., Last, P.R. & Stuart-Smith, R.

Reviewer(s): Ralph, G.

**Facilitator(s) and
Compiler(s):** Linardich, C.

Bibliography

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External Resources

For [Supplementary Material](#), and for [Images and External Links to Additional Information](#), please see the Red List website.

Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
9. Marine Neritic -> 9.2. Marine Neritic - Subtidal Rock and Rocky Reefs	Resident	Suitable	Yes
9. Marine Neritic -> 9.3. Marine Neritic - Subtidal Loose Rock/pebble/gravel	Resident	Suitable	Yes
9. Marine Neritic -> 9.4. Marine Neritic - Subtidal Sandy	Resident	Suitable	Yes
9. Marine Neritic -> 9.5. Marine Neritic - Subtidal Sandy-Mud	Resident	Suitable	Yes
9. Marine Neritic -> 9.6. Marine Neritic - Subtidal Muddy	Resident	Suitable	Yes
9. Marine Neritic -> 9.7. Marine Neritic - Macroalgal/Kelp	Resident	Suitable	Yes

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity	Impact Score
5. Biological resource use -> 5.4. Fishing & harvesting aquatic resources -> 5.4.4. Unintentional effects: (large scale) [harvest]	Past, unlikely to return	Whole (>90%)	Rapid declines	Past impact
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation 2. Species Stresses -> 2.1. Species mortality		

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action in Place
In-place land/water protection
Occurs in at least one protected area: No

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed
1. Research -> 1.2. Population size, distribution & trends
1. Research -> 1.3. Life history & ecology
2. Conservation Planning -> 2.1. Species Action/Recovery Plan
3. Monitoring -> 3.1. Population trends

Additional Data Fields

Distribution
Estimated area of occupancy (AOO) (km ²): 12
Continuing decline in area of occupancy (AOO): Unknown
Estimated extent of occurrence (EOO) (km ²): 1262
Continuing decline in extent of occurrence (EOO): Unknown
Lower depth limit (m): 38
Upper depth limit (m): 15
Population
Continuing decline of mature individuals: Yes
Population severely fragmented: Yes
Continuing decline in subpopulations: Unknown
Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Unknown

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